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SEAT No. :

PC-372

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[6361]-224

B.E. (Mechanical) (Insem.) (Honors)
MODELLING AND SIMULATION OF EHV
(2019 Pattern) (Semester - VII) (302034MJ)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4*
- 2) *Neat Diagrams must be drawn wherever necessary.*
- 3) *Figure to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Define motor drives for electric vehicles with its advantage and requirements. [8]

b) Explain Basic Elements of the Electric Drive Systems with neat sketch. [6]

OR

Q2) a) Classify EV motor drives with its comparative analysis. [6]

b) Explain DC motor drives with its working principal, advantages, disadvantages and applications. [8]

Q3) a) Define energy storage system, explain its classifications. [8]

b) Explain the constructional details of li-ion battery with its advantages, disadvantages and applications. [8]

OR

Q4) a) Compare types of battery cooling with its significant features. [8]

b) Explain NVH analysis with its importance in electric vehicles. [8]

